Laws in spherical geometry

In[1]:=
```
SpherSinLaw[a_, alpha_, b_, beta_] = Sin[a] Sin[beta] == Sin[b] Sin[alpha];
                                     Sinus  Sinus          Sinus  Sinus

SpherCosLaw[a_, b_, c_, alpha_] = Cos[a] == Cos[b] Cos[c] + Sin[b] Sin[c] Cos[alpha];
                                  Kosinus  Kosinus Kosinus  Sinus  Sinus  Kosinus

SquaredSpherCosLaw[a_, b_, c_, alpha_] =
  1 - Sin[a]^2 == (Cos[b] Cos[c] + Sin[b] Sin[c] Cos[alpha])^2;
      Sinus        Kosinus Kosinus  Sinus  Sinus  Kosinus
```

C-variable definitions

In[4]:=
```
replaceVarRule = {
  Sin[a]^2 → SinaSq, Sin[b]^2 → SinbSq, Sin[c]^2 → SincSq,
  Csc[a] → 1 / Sqrt[SinaSq], Csc[b] → 1 / Sqrt[SinbSq], Csc[c] → 1 / Sqrt[SincSq],
  Sin[a / 2]^2 → ScaledSina2Sq / 4,
  Sin[b / 2]^2 → ScaledSinb2Sq / 4, Sin[c / 2]^2 → ScaledSinc2Sq / 4,
  Sin[a / 2]^4 → ScaledSina2Sq^2 / 16,
  Sin[b / 2]^4 → ScaledSinb2Sq^2 / 16, Sin[c / 2]^4 → ScaledSinc2Sq^2 / 16,
  Cos[alpha] → CosAlpha, Cos[beta] → CosBeta, Cos[gamma] → CosGamma,
  Sin[ma]^2 → SinmaSq, Sin[mb]^2 → SinmbSq, Sin[mc]^2 → SinmcSq,
  Cos[alphamab] → Cosalphamab, Cos[betamab] → Cosbetamab,
  Cos[alphamac] → Cosalphamac, Cos[gammamac] → Cosgammamac,
  Cos[betambc] → Cosbetambc, Cos[gammambc] → Cosgammambc,
  Cos[alphammm] → Cosalphammm,
  Cos[betammm] → Cosbetammm, Cos[gammammm] → Cosgammammm,
  Csc[a / 2] → 2 / Sqrt[ScaledSina2Sq],
  Csc[b / 2] → 2 / Sqrt[ScaledSinb2Sq], Csc[c / 2] → 2 / Sqrt[ScaledSinc2Sq],
  Sec[a / 2] → 2 / Sqrt[4 - ScaledSina2Sq],
  Sec[b / 2] → 2 / Sqrt[4 - ScaledSinb2Sq], Sec[c / 2] → 2 / Sqrt[4 - ScaledSinc2Sq],
  Sin[a] → Sqrt[SinaSq], Sin[b] → Sqrt[SinbSq], Sin[c] → Sqrt[SincSq],
  Cos[alphasab] → Cosalphasab, Cos[alphasac] → Cosalphasac,
  Cos[betasba] → Cosbetasba, Cos[betasbc] → Cosbetasbc,
  Cos[gammasca] → Cosgammasca, Cos[gammascb] → Cosgammascb,
  Cos[alphasbb] → Cosalphasbb, Cos[alphascc] → Cosalphascc,
  Cos[betasaa] → Cosbetasaa, Cos[betascc] → Cosbetascc,
  Cos[gammasaa] → Cosgammasaa, Cos[gammasbb] → Cosgammasbb
 };
```

In[5]:=
```
inverseReplaceVarRule = {
   SinaSq → Sin[a]^2, SinbSq → Sin[b]^2, SincSq → Sin[c]^2,
            Sinus              Sinus              Sinus
   ScaledSina2Sq → 4 Sin[a / 2]^2,
                     Sinus
   ScaledSinb2Sq → 4 Sin[b / 2]^2, ScaledSinc2Sq → 4 Sin[c / 2]^2,
                     Sinus                           Sinus
   CosAlpha → Cos[alpha], CosBeta → Cos[beta], CosGamma → Cos[gamma],
              Kosinus             Kosinus              Kosinus
   SinmaSq → Sin[ma]^2, SinmbSq → Sin[mb]^2, SinmcSq → Sin[mc]^2,
             Sinus              Sinus              Sinus
   Cosalphamab → Cos[alphamab], Cosbetamab → Cos[betamab], Cosalphamac → Cos[alphamac],
                 Kosinus                     Kosinus                     Kosinus
   Cosgammamac → Cos[gammamac], Cosbetambc → Cos[betambc], Cosgammambc → Cos[gammambc],
                 Kosinus                     Kosinus                     Kosinus
   Cosalphammm → Cos[alphammm], Cosbetammm → Cos[betammm], Cosgammammm → Cos[gammammm],
                 Kosinus                     Kosinus                     Kosinus
   Cosalphasab → Cos[alphasab], Cosalphasac → Cos[alphasac],
                 Kosinus                     Kosinus
   Cosbetasba → Cos[betasba], Cosbetasbc → Cos[betasbc],
                Kosinus                    Kosinus
   Cosgammasca → Cos[gammasca], Cosgammascb → Cos[gammascb],
                 Kosinus                     Kosinus
   Cosalphasbb → Cos[alphasbb], Cosalphascc → Cos[alphascc],
                 Kosinus                     Kosinus
   Cosbetasaa → Cos[betasaa], Cosbetascc → Cos[betascc],
                Kosinus                    Kosinus
   Cosgammasaa → Cos[gammasaa], Cosgammasbb → Cos[gammasbb]
                 Kosinus                     Kosinus
  };
```

Check consistency of variable Definitions

In[6]:=
```
And @@ (Simplify[Equal @@@ (replaceVarRule /. inverseReplaceVarRule),
und     vereinfache gleich
   {a ≥ 0, a ≤ Pi / 2, b ≥ 0, b ≤ Pi / 2, c ≥ 0, c ≤ Pi / 2}])
                Kreiszahl π        Kreiszahl π        Kreiszahl π
```

Out[6]= True

In[7]:=
```
sdeVarNameRule = {
   Sin[sa]^2 → SinsaSq, Sin[sb]^2 → SinsbSq, Sin[sc]^2 → SinscSq,
   Sinus                Sinus                Sinus
   Sin[ea]^2 → SineaSq, Sin[eb]^2 → SinebSq, Sin[ec]^2 → SinecSq,
   Sinus                Sinus                Sinus
   Sin[da]^2 → SindaSq, Sin[db]^2 → SindbSq, Sin[dc]^2 → SindcSq};
   Sinus                Sinus                Sinus
```

Basic replacement for Sec, Cos of half angle, etc.: Sec=1/Cos, Csc=1/Sin, Cos[a]=1-2Sin[a/2]^2, Sin[a]=2 Sin[a/2]Cos[a/2]

In[8]:=
```
basicRule = {
  Cos[2 a] → 1 - 2 Sin[a]^2, Cos[2 b] → 1 - 2 Sin[b]^2, Cos[2 c] → 1 - 2 Sin[c]^2,
  Kosinus        Sinus        Kosinus        Sinus        Kosinus        Sinus
  Cos[a]^2 → 1 - Sin[a]^2, Cos[b]^2 → 1 - Sin[b]^2, Cos[c]^2 → 1 - Sin[c]^2,
  Kosinus        Sinus        Kosinus        Sinus        Kosinus        Sinus
  Sec[a] → 1 / (1 - 2 Sin[a / 2]^2),
  Sekante        Sinus
  Sec[b] → 1 / (1 - 2 Sin[b / 2]^2), Sec[c] → 1 / (1 - 2 Sin[c / 2]^2),
  Sekante        Sinus              Sekante        Sinus
  Sec[a / 2]^2 → 1 / (1 - Sin[a / 2]^2),
  Sekante            Sinus
  Sec[b / 2]^2 → 1 / (1 - Sin[b / 2]^2), Sec[c / 2]^2 → 1 / (1 - Sin[c / 2]^2),
  Sekante            Sinus              Sekante            Sinus
  Cos[a] → (1 - 2 Sin[a / 2]^2),
  Kosinus      Sinus
  Cos[b] → (1 - 2 Sin[b / 2]^2), Cos[c] → (1 - 2 Sin[c / 2]^2),
  Kosinus      Sinus              Kosinus      Sinus
  Cot[a] → (1 - 2 Sin[a / 2]^2) / Sin[a],
  Kotangens    Sinus              Sinus
  Cot[b] → (1 - 2 Sin[b / 2]^2) / Sin[b], Cot[c] → (1 - 2 Sin[c / 2]^2) / Sin[c],
  Kotangens    Sinus              Sinus    Kotangens    Sinus              Sinus
  Cot[a / 2]^2 → (1 - Sin[a / 2]^2) / Sin[a / 2]^2, Cot[b / 2]^2 →
  Kotangens        Sinus              Sinus              Kotangens
   (1 - Sin[b / 2]^2) / Sin[b / 2]^2, Cot[c / 2]^2 → (1 - Sin[c / 2]^2) / Sin[c / 2]^2,
       Sinus              Sinus              Kotangens              Sinus              Sinus
  Sin[x] → 2 Sin[x / 2] Cos[x / 2],
  Sinus      Sinus      Kosinus
  Sin[2 a] → 2 Sin[a] Cos[a], Sin[2 b] → 2 Sin[b] Cos[b], Sin[2 c] → 2 Sin[c] Cos[c],
  Sinus      Sinus    Kosinus  Sinus      Sinus    Kosinus  Sinus      Sinus    Kosinus
  Sin[alpha]^2 → 1 - Cos[alpha]^2,
  Sinus            Kosinus
  Sin[beta]^2 → 1 - Cos[beta]^2, Sin[gamma]^2 → 1 - Cos[gamma]^2,
  Sinus            Kosinus        Sinus            Kosinus
  Cos[a - b] → Cos[a] Cos[b] + Sin[a] Sin[b],
  Kosinus      Kosinus Kosinus   Sinus   Sinus
  Cos[a - c] → Cos[a] Cos[c] + Sin[a] Sin[c], Cos[b - c] → Cos[b] Cos[c] + Sin[b] Sin[c],
  Kosinus      Kosinus Kosinus   Sinus   Sinus   Kosinus      Kosinus Kosinus   Sinus   Sinus
  Cos[a + b] → Cos[a] Cos[b] - Sin[a] Sin[b],
  Kosinus      Kosinus Kosinus   Sinus   Sinus
  Cos[a + c] → Cos[a] Cos[c] - Sin[a] Sin[c], Cos[b + c] → Cos[b] Cos[c] - Sin[b] Sin[c]
  Kosinus      Kosinus Kosinus   Sinus   Sinus   Kosinus      Kosinus Kosinus   Sinus   Sinus
 };
```

In[9]:=
```
And @@ (Equal @@@ basicRule) // Simplify
und    gleich                  vereinfache
```

Out[9]=
```
True
```

In[10]:=
```
basicRule2 = Join[basicRule, {
                 verknüpfe
    Csc[a] → 1 / (2 Sin[a / 2] Cos[a / 2]),
    Kosekans      Sinus      Kosinus
    Csc[b] → 1 / (2 Sin[b / 2] Cos[b / 2]), Csc[c] → 1 / (2 Sin[c / 2] Cos[c / 2])
    Kosekans      Sinus      Kosinus      Kosekans      Sinus      Kosinus
   }];
```

In[11]:=
```
And @@ (Equal @@@ basicRule2) // Simplify
und    gleich                  vereinfache
```

Out[11]=
```
True
```

Formula to calculate half edges:
4 Sin[a/2]^2 / Sin[a]^2 = 2 / (1+Sqrt[1-Sin[a]^2)

In[12]:=
```
halfEdgeRule = {
   Sin[a / 2] ^2 → Sin[a] ^2 / (2 (Sqrt[1 - Sin[a] ^2] + 1)),
   Sinus           Sinus          Quadra···  Sinus
   Sin[b / 2] ^2 → Sin[b] ^2 / (2 (Sqrt[1 - Sin[b] ^2] + 1)),
   Sinus           Sinus          Quadra···  Sinus
   Sin[c / 2] ^2 → Sin[c] ^2 / (2 (Sqrt[1 - Sin[c] ^2] + 1))};
   Sinus           Sinus          Quadra···  Sinus
```

In[13]:=
```
Simplify[Equal @@@ halfEdgeRule, {a ≥ 0, a ≤ Pi / 2, b ≥ 0, b ≤ Pi / 2, c ≥ 0, c ≤ Pi / 2}]
vereinfache gleich                          Kreiszahl π     Kreiszahl π     Kreiszahl
```

Out[13]=
```
{True, True, True}
```

In[14]:=
```
formulas = halfEdgeRule /. replaceVarRule /. Rule[a_, b_] → Rule[4 a, 4 b]
                                             Regel          Regel
```

Out[14]=

$$\left\{\text{ScaledSina2Sq} \to \frac{2\,\text{SinaSq}}{1+\sqrt{1-\text{SinaSq}}},\ \text{ScaledSinb2Sq} \to \frac{2\,\text{SinbSq}}{1+\sqrt{1-\text{SinbSq}}},\ \text{ScaledSinc2Sq} \to \frac{2\,\text{SincSq}}{1+\sqrt{1-\text{SincSq}}}\right\}$$

Spherical triangle with edges a, b, c: calculate interior angles alpha, beta, gamma

In[15]:=
```
CosOrigAngSol = (Solve[{
                 löse
        SpherCosLaw[a, b, c, alpha],
        SpherCosLaw[b, a, c, beta],
        SpherCosLaw[c, a, b, gamma]}, {Cos[alpha], Cos[beta], Cos[gamma]}] // Simplify)[[
                                       Kosinus     Kosinus     Kosinus       vereinfache
   1]];
```

In[16]:=
```
SinOrigAngSol = Solve[{
                 löse
      SpherSinLaw[a, alpha, b, beta],
      SpherSinLaw[c, gamma, b, beta],
      SpherSinLaw[a, alpha, c, gamma]
     }, {Sin[beta], Sin[gamma]}][[1]];
        Sinus        Sinus
```

In[17]:=
```
formulas = Join[formulas, (CosOrigAngSol /. basicRule) /. replaceVarRule // Simplify]
           verknüpfe                                                      vereinfache
```

Out[17]=

$$\left\{\text{ScaledSina2Sq} \to \frac{2\,\text{SinaSq}}{1+\sqrt{1-\text{SinaSq}}},\ \text{ScaledSinb2Sq} \to \frac{2\,\text{SinbSq}}{1+\sqrt{1-\text{SinbSq}}},\ \text{ScaledSinc2Sq} \to \frac{2\,\text{SincSq}}{1+\sqrt{1-\text{SincSq}}},\right.$$

$$\text{CosAlpha} \to \frac{-2\,\text{ScaledSina2Sq} - \text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq}) + 2\,\text{ScaledSinc2Sq}}{4\sqrt{\text{SinbSq}}\,\sqrt{\text{SincSq}}},$$

$$\text{CosBeta} \to \frac{-2\,\text{ScaledSinb2Sq} - \text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq}) + 2\,\text{ScaledSinc2Sq}}{4\sqrt{\text{SinaSq}}\,\sqrt{\text{SincSq}}},$$

$$\text{CosGamma} \to \left.\frac{2\,\text{ScaledSina2Sq} + 2\,\text{ScaledSinb2Sq} - \text{ScaledSina2Sq}\,\text{ScaledSinb2Sq} - 2\,\text{ScaledSinc2Sq}}{4\sqrt{\text{SinaSq}}\,\sqrt{\text{SinbSq}}}\right\}$$

Calculation of the length of side bisections sa, sb, sc

In[18]:=
```
CosSSol = Solve[{
          löse
      SpherCosLaw[sa, a / 2, b, gamma],
      SpherCosLaw[sb, b / 2, c, alpha],
      SpherCosLaw[sc, c / 2, a, beta]}, {Cos[sa], Cos[sb], Cos[sc]}][[1]];
                                         Kosinus   Kosinus   Kosinus
```

When solving the squared cosine law then for each equation there is the positive and negative solution, i . e . in the end there are eight solutions. All eight solutions are the same besides the sign.

In[19]:=
```
SinSSol = Solve[{
          löse
       SquaredSpherCosLaw[sa, a / 2, b, gamma],
       SquaredSpherCosLaw[sb, b / 2, c, alpha],
       SquaredSpherCosLaw[sc, c / 2, a, beta]}, {Sin[sa], Sin[sb], Sin[sc]}][[8]] //
                                                  Sinus     Sinus     Sinus
   Simplify;
   vereinfache
```

In[20]:=
```
SSol = (Solve[{
        löse
          SpherCosLaw[sa, a / 2, b, gamma],
          SpherCosLaw[sb, b / 2, c, alpha],
          SpherCosLaw[sc, c / 2, a, beta]}, {sa, sb, sc}][[8]] /.
      {c1 → 0, c2 → 0, c3 → 0} // Simplify);
                                  vereinfache
```

Formulas for sa, sb, sc

In[21]:=
```
sFormula = Map[Simplify[Rule[# /. sdeVarNameRule, # /. SinSSol /. CosOrigAngSol]] //.
            w···  vereinfache Regel
      Join[basicRule, replaceVarRule] &,
      verknüpfe
    {Sin[sa]^2, Sin[sb]^2, Sin[sc]^2}] // FullSimplify;
     Sinus       Sinus       Sinus          vereinfache vollständig
```

Check that the results are unchanged when swapping variable names

In[22]:=
```
{(Sin[sa]^2 //. Join[sFormula, sdeVarNameRule]) -
  Sinus          verknüpfe
   (Sin[sb]^2 //. Join[sFormula, sdeVarNameRule] /.
    Sinus          verknüpfe
     {SinaSq → SinbSq, SinbSq → SinaSq,
      ScaledSina2Sq → ScaledSinb2Sq, ScaledSinb2Sq → ScaledSina2Sq}),
  (Sin[sa]^2 //. Join[sFormula, sdeVarNameRule]) -
   Sinus          verknüpfe
   (Sin[sc]^2 //. Join[sFormula, sdeVarNameRule] /.
    Sinus          verknüpfe
     {SinaSq → SincSq, SincSq → SinaSq,
      ScaledSina2Sq → ScaledSinc2Sq, ScaledSinc2Sq → ScaledSina2Sq}),
  (Sin[sb]^2 //. Join[sFormula, sdeVarNameRule]) -
   Sinus          verknüpfe
   (Sin[sc]^2 //. Join[sFormula, sdeVarNameRule] /.
    Sinus          verknüpfe
     {SinbSq → SincSq, SincSq → SinbSq,
      ScaledSinb2Sq → ScaledSinc2Sq, ScaledSinc2Sq → ScaledSinb2Sq})} // Simplify
                                                                        vereinfache
```

Out[22]=
```
{0, 0, 0}
```

In[23]:=
```
formulas = Join[formulas, sFormula];
           verknüpfe
```

Calculate the lengths of the connection between two side bisection points: ma, mb, mc

Use previous solution of interior angles to solve for Cos[m]

In[24]:=
```
CosMSol = (Solve[{
           löse
        SpherCosLaw[ma, b / 2, c / 2, alpha],
        SpherCosLaw[mb, a / 2, c / 2, beta],
        SpherCosLaw[mc, a / 2, b / 2, gamma]} /. CosOrigAngSol,
      {Cos[ma], Cos[mb], Cos[mc]}] // Simplify)[[1]];
       Kosinus   Kosinus   Kosinus     vereinfache
```

Use previous solution of interior angles to solve for Sin[m]:

When solving the squared cosine law then for each equation there is the positive and negative solution, i . e . in the end there are eight solutions. All eight solutions are the same besides the sign.

In[25]:=
```
SinMSol = (Solve[{
            löse
          SquaredSpherCosLaw[ma, b / 2, c / 2, alpha],
          SquaredSpherCosLaw[mb, a / 2, c / 2, beta],
          SquaredSpherCosLaw[mc, a / 2, b / 2, gamma]} /. CosOrigAngSol,
        {Sin[ma], Sin[mb], Sin[mc]}] // Simplify)[[8]];
         Sinus     Sinus     Sinus          vereinfache
```

Verify that the two solutions are the same

In[26]:=
```
{1 - Sin[ma]^2 - Cos[ma]^2, 1 - Sin[mb]^2 - Cos[mb]^2, 1 - Sin[mc]^2 - Cos[mc]^2} /.
    Sinus        Kosinus        Sinus        Kosinus        Sinus        Kosinus
  Join[CosMSol, SinMSol] // Simplify
  verknüpfe                 vereinfache
```

Out[26]=
```
{0, 0, 0}
```

Formulas for Sin^2 ma, Sin^2 mb, Sin^2 mc

In[27]:=
```
{
   Sin[ma]^2 → (Sin[ma]^2 /. SinMSol),
   Sinus        Sinus
   Sin[mb]^2 → (Sin[mb]^2 /. SinMSol),
   Sinus        Sinus
   Sin[mc]^2 → (Sin[mc]^2 /. SinMSol),
   Sinus        Sinus
   4 Sin[ma]^2 → (4 Sin[ma]^2 /. SinMSol),
     Sinus          Sinus
   4 Sin[mb]^2 → (4 Sin[mb]^2 /. SinMSol),
     Sinus          Sinus
   4 Sin[mc]^2 → (4 Sin[mc]^2 /. SinMSol)} // ExpandAll;
     Sinus          Sinus                   multipliziere alle aus
```

In[28]:=
```
mFormula = (% //. basicRule // ExpandAll) /. replaceVarRule // FullSimplify;
                   multipliziere alle aus                      vereinfache vollstär
```

Check that the results are unchanged when swapping variable names

In[29]:=
```
{
  (SinmaSq - (SinmbSq /. mFormula /. {SinaSq → SinbSq, SinbSq → SinaSq,
        ScaledSina2Sq → ScaledSinb2Sq, ScaledSinb2Sq → ScaledSina2Sq})) /. mFormula,
  (SinmaSq - (SinmcSq /. mFormula /. {SinaSq → SincSq, SincSq → SinaSq,
        ScaledSina2Sq → ScaledSinc2Sq, ScaledSinc2Sq → ScaledSina2Sq})) /. mFormula,
  (SinmbSq - (SinmcSq /. mFormula /. {SinbSq → SincSq, SincSq → SinbSq,
        ScaledSinc2Sq → ScaledSinb2Sq, ScaledSinb2Sq → ScaledSinc2Sq})) /. mFormula
 } // Simplify
      vereinfache
```

Out[29]=
```
{0, 0, 0}
```

In[30]:=
```
formulas = Join[formulas, mFormula];
           verknüpfe
```

sa bisects ma into two pieces mab & mac; it bisects alpha into two angles

alphasab & alphasac.
Their ratios are (use spherical sine laws to prove)
Sin[mab]/Sin[mac] = Cos[c/2]/Cos[b/2]
Sin[alphasab]/Sin[alphasac] = Sin[c]/Sin[b] = Sin[gamma]/Sin[beta]

In[31]:=
```
bisectRatioRule = Solve[{
                  löse
        SpherSinLaw[mab, alphasab, saH, gammambc],
        SpherSinLaw[mac, alphasac, saH, betambc],
        SpherSinLaw[c / 2, gammambc, b / 2, betambc],
        SpherSinLaw[b, betasaa, a / 2, alphasab],
        SpherSinLaw[c, gammasaa, a / 2, alphasac],
        Sin[betasaa] == Sin[Pi - gammasaa]
        Sinus          Si···  Kreiszahl π
       } // Simplify,
            vereinfache
      {Sin[mab], Sin[alphasab], Sin[gammambc], Sin[betasaa], Sin[gammasaa], Sin[saH]}][[
       Sinus      Sinus          Sinus          Sinus          Sinus          Sinus
     1]] // Simplify;
            vereinfache
```

In[32]:=
```
{
  Sin[mab] / Sin[mac],
  Sinus      Sinus
  Sin[alphasab] / Sin[alphasac]
  Sinus           Sinus
 } /. bisectRatioRule
```

Out[32]=

$$\left\{\text{Cos}\left[\frac{c}{2}\right]\text{Sec}\left[\frac{b}{2}\right],\ \text{Csc}[b]\ \text{Sin}[c]\right\}$$

Calculation of interior angles of the refined triangles: alphamab, betamab, alphamac, gammamac, betambc, gammambc, alphammm, betammm, gammammm

In[33]:=
```
refinedAngSol = Solve[{
                löse
        SpherCosLaw[a / 2, b / 2, mc, alphamab],
        SpherCosLaw[b / 2, a / 2, mc, betamab],
        SpherCosLaw[a / 2, c / 2, mb, alphamac],
        SpherCosLaw[c / 2, a / 2, mb, gammamac],
        SpherCosLaw[b / 2, c / 2, ma, betambc],
        SpherCosLaw[c / 2, b / 2, ma, gammambc],
        SpherCosLaw[ma, mb, mc, alphammm],
        SpherCosLaw[mb, ma, mc, betammm],
        SpherCosLaw[mc, ma, mb, gammammm]} /. Join[CosMSol, SinMSol],
                                              verknüpfe
      {Cos[alphamab], Cos[betamab], Cos[alphamac], Cos[gammamac], Cos[betambc],
       Kosinus        Kosinus        Kosinus        Kosinus        Kosinus
       Cos[gammambc], Cos[alphammm], Cos[betammm], Cos[gammammm]}][[1]] // Simplify;
       Kosinus        Kosinus        Kosinus        Kosinus                vereinfache
```

Check that the results are unchanged when swapping variable names

In[34]:=
```
{(Cos[alphammm] - ((Cos[betammm] /. refinedAngSol) /. {a → b, b → a})) /. refinedAngSol,
  Kosinus                Kosinus
 (Cos[alphammm] - ((Cos[gammammm] /. refinedAngSol) /. {a → c, c → a})) /. refinedAngSol,
  Kosinus                Kosinus
 (Cos[alphamab] - ((Cos[betamab] /. refinedAngSol) /. {a → b, b → a})) /. refinedAngSol,
  Kosinus                Kosinus
 (Cos[alphamac] - ((Cos[gammamac] /. refinedAngSol) /. {a → c, c → a})) /. refinedAngSol,
  Kosinus                Kosinus
 (Cos[betambc] - ((Cos[gammambc] /. refinedAngSol) /. {b → c, c → b})) /. refinedAngSol,
  Kosinus                Kosinus
 (Cos[alphamab] - ((Cos[gammamac] /. refinedAngSol) /. {a → b, b → c, c → a})) /.
  Kosinus                Kosinus
  refinedAngSol,
 (Cos[betamab] - ((Cos[betambc] /. refinedAngSol) /. {a → c, b → b, c → a})) /.
  Kosinus                Kosinus
  refinedAngSol} // Simplify
                    vereinfache
```

Out[34]=
```
{0, 0, 0, 0, 0, 0, 0}
```

In[35]:=
```
(refinedAngSol // Simplify) /. basicRule /. replaceVarRule // FullSimplify
                vereinfache                                   vereinfache vollständig
```

Out[35]=
```
{Cosalphamab →
  (-ScaledSina2Sq + ScaledSinb2Sq + ScaledSinc2Sq) / (√(8 - 2 ScaledSina2Sq) √ScaledSinb2Sq
     √((ScaledSina2Sq (-2 + ScaledSinb2Sq - ScaledSinc2Sq) - ScaledSinb2Sq
           (2 + ScaledSinc2Sq) + 2 (3 ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
        ((-4 + ScaledSina2Sq) (-4 + ScaledSinb2Sq)))),
 Cosbetamab → (ScaledSina2Sq - ScaledSinb2Sq + ScaledSinc2Sq) /
   (√2 √ScaledSina2Sq √(4 - ScaledSinb2Sq
     √((ScaledSina2Sq (-2 + ScaledSinb2Sq - ScaledSinc2Sq) - ScaledSinb2Sq
           (2 + ScaledSinc2Sq) + 2 (3 ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
        ((-4 + ScaledSina2Sq) (-4 + ScaledSinb2Sq)))),
 Cosalphamac → (-ScaledSina2Sq + ScaledSinb2Sq + ScaledSinc2Sq) /
   (√(8 - 2 ScaledSina2Sq) √ScaledSinc2Sq
     √((-ScaledSinb2Sq (-6 + ScaledSinc2Sq) + ScaledSina2Sq (-2 - ScaledSinb2Sq +
           ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
        ((-4 + ScaledSina2Sq) (-4 + ScaledSinc2Sq)))),
 Cosgammamac → (ScaledSina2Sq + ScaledSinb2Sq - ScaledSinc2Sq) /
   (√2 √ScaledSina2Sq √(4 - ScaledSinc2Sq
     √((-ScaledSinb2Sq (-6 + ScaledSinc2Sq) + ScaledSina2Sq (-2 - ScaledSinb2Sq +
           ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
        ((-4 + ScaledSina2Sq) (-4 + ScaledSinc2Sq)))),
 Cosbetambc → (ScaledSina2Sq - ScaledSinb2Sq + ScaledSinc2Sq) /
   (√(8 - 2 ScaledSinb2Sq) √ScaledSinc2Sq
     √((ScaledSinb2Sq (-2 + ScaledSinc2Sq) - ScaledSina2Sq (-6 + ScaledSinb2Sq +
           ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
        ((-4 + ScaledSinb2Sq) (-4 + ScaledSinc2Sq)))),
```

```
Cosgammambc → (ScaledSina2Sq + ScaledSinb2Sq - ScaledSinc2Sq) /
  (√2 √ScaledSinb2Sq √(4 - ScaledSinc2Sq)
    √((ScaledSinb2Sq (-2 + ScaledSinc2Sq) - ScaledSina2Sq (-6 + ScaledSinb2Sq +
          ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSinb2Sq) (-4 + ScaledSinc2Sq)))),
Cosalphammm → ((-ScaledSina2Sq + ScaledSinb2Sq + ScaledSinc2Sq)
    (-8 + ScaledSina2Sq + ScaledSinb2Sq + ScaledSinc2Sq)) /
  (2 (-4 + ScaledSina2Sq) √(4 - ScaledSinb2Sq) √(4 - ScaledSinc2Sq)
    √((-ScaledSinb2Sq (-6 + ScaledSinc2Sq) + ScaledSina2Sq (-2 - ScaledSinb2Sq +
          ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSina2Sq) (-4 + ScaledSinc2Sq)))
    √((ScaledSina2Sq (-2 + ScaledSinb2Sq - ScaledSinc2Sq) - ScaledSinb2Sq
          (2 + ScaledSinc2Sq) + 2 (3 ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSina2Sq) (-4 + ScaledSinb2Sq)))),
Cosbetammm → ((ScaledSina2Sq - ScaledSinb2Sq + ScaledSinc2Sq)
    (-8 + ScaledSina2Sq + ScaledSinb2Sq + ScaledSinc2Sq)) /
  (2 √(4 - ScaledSina2Sq) (-4 + ScaledSinb2Sq) √(4 - ScaledSinc2Sq)
    √((ScaledSinb2Sq (-2 + ScaledSinc2Sq) - ScaledSina2Sq (-6 + ScaledSinb2Sq +
          ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSinb2Sq) (-4 + ScaledSinc2Sq)))
    √((ScaledSina2Sq (-2 + ScaledSinb2Sq - ScaledSinc2Sq) - ScaledSinb2Sq
          (2 + ScaledSinc2Sq) + 2 (3 ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSina2Sq) (-4 + ScaledSinb2Sq)))),
Cosgammammm → ((ScaledSina2Sq + ScaledSinb2Sq - ScaledSinc2Sq)
    (-8 + ScaledSina2Sq + ScaledSinb2Sq + ScaledSinc2Sq)) /
  (2 √(4 - ScaledSina2Sq) √(4 - ScaledSinb2Sq) (-4 + ScaledSinc2Sq)
    √((-ScaledSinb2Sq (-6 + ScaledSinc2Sq) + ScaledSina2Sq (-2 - ScaledSinb2Sq +
          ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSina2Sq) (-4 + ScaledSinc2Sq)))
    √((ScaledSinb2Sq (-2 + ScaledSinc2Sq) - ScaledSina2Sq (-6 + ScaledSinb2Sq +
          ScaledSinc2Sq) + 2 (-ScaledSinc2Sq + SinaSq + SinbSq + SincSq)) /
       ((-4 + ScaledSinb2Sq) (-4 + ScaledSinc2Sq))))}
```

Angles inside triangles built by [side bisection, original edge and half original edge]: alphasab, alphasac, betasba, betasbc, gammasca, gammascb; alphasbb, alphascc, betasaa, betascc, gammasaa, gammasbb

First way : use spherical sine law

In[36]:=
```
SinBisectAngSol = (Solve[{
                   löse
        SpherSinLaw[a / 2, alphasab, sa, gamma],
        SpherSinLaw[a / 2, alphasac, sa, beta],
        SpherSinLaw[b, betasaa, sa, gamma],
        SpherSinLaw[c, gammasaa, sa, beta],
        SpherSinLaw[b / 2, betasbc, sb, alpha],
        SpherSinLaw[b / 2, betasba, sb, gamma],
        SpherSinLaw[a, alphasbb, sb, gamma],
        SpherSinLaw[c, gammasbb, sb, alpha],
        SpherSinLaw[c / 2, gammascb, sc, alpha],
        SpherSinLaw[c / 2, gammasca, sc, beta],
        SpherSinLaw[a, alphascc, sc, beta],
        SpherSinLaw[b, betascc, sc, alpha]} /. SinSSol,
      {Sin[alphasab], Sin[alphasac],
       Sinus          Sinus
       Sin[betasba], Sin[betasbc], Sin[gammasca], Sin[gammascb],
       Sinus         Sinus         Sinus          Sinus
       Sin[alphasbb], Sin[alphascc], Sin[betasaa],
       Sinus          Sinus          Sinus
       Sin[betascc], Sin[gammasaa], Sin[gammasbb]}] // Simplify)[[1]];
       Sinus         Sinus          Sinus           vereinfache
```

Second way: use spherical cosine law

In[37]:=
```
CosBisectAngSol = (Solve[{
                   löse
        SpherCosLaw[a / 2, b, sa, alphasab],
        SpherCosLaw[a / 2, c, sa, alphasac],
        SpherCosLaw[b / 2, a, sb, betasba],
        SpherCosLaw[b / 2, c, sb, betasbc],
        SpherCosLaw[c / 2, a, sc, gammasca],
        SpherCosLaw[c / 2, b, sc, gammascb],
        SpherCosLaw[a, b / 2, sb, alphasbb],
        SpherCosLaw[c, b / 2, sb, gammasbb],
        SpherCosLaw[a, c / 2, sc, alphascc],
        SpherCosLaw[b, c / 2, sc, betascc],
        SpherCosLaw[b, a / 2, sa, betasaa],
        SpherCosLaw[c, a / 2, sa, gammasaa]} /. Join[SinSSol, CosSSol], {Cos[alphasab],
                                                verknüpfe                 Kosinus
       Cos[alphasac], Cos[betasba], Cos[betasbc], Cos[gammasca], Cos[gammascb],
       Kosinus        Kosinus       Kosinus        Kosinus        Kosinus
       Cos[alphasbb], Cos[alphascc], Cos[betasaa],
       Kosinus        Kosinus        Kosinus
       Cos[betascc], Cos[gammasaa], Cos[gammasbb]}] // Simplify)[[1]];
       Kosinus       Kosinus        Kosinus          vereinfache
```

Confirm that the two solutions are the same

In[38]:=
```
{(Cos[alphasab]^2 /. CosBisectAngSol) - (1 - Sin[alphasab]^2 /. SinBisectAngSol),
  Kosinus                                  Sinus
   (Cos[alphasac]^2 /. CosBisectAngSol) - (1 - Sin[alphasac]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[betasba]^2 /. CosBisectAngSol) - (1 - Sin[betasba]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[betasbc]^2 /. CosBisectAngSol) - (1 - Sin[betasbc]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[gammasca]^2 /. CosBisectAngSol) - (1 - Sin[gammasca]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[gammascb]^2 /. CosBisectAngSol) - (1 - Sin[gammascb]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[alphasbb]^2 /. CosBisectAngSol) - (1 - Sin[alphasbb]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[alphascc]^2 /. CosBisectAngSol) - (1 - Sin[alphascc]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[betasaa]^2 /. CosBisectAngSol) - (1 - Sin[betasaa]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[betascc]^2 /. CosBisectAngSol) - (1 - Sin[betascc]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[gammasaa]^2 /. CosBisectAngSol) - (1 - Sin[gammasaa]^2 /. SinBisectAngSol),
    Kosinus                                   Sinus
   (Cos[gammasbb]^2 /. CosBisectAngSol) - (1 - Sin[gammasbb]^2 /. SinBisectAngSol)} //.
    Kosinus                                   Sinus
  Join[basicRule, CosOrigAngSol] // Simplify
  verknüpfe                         vereinfache
```

Out[38]=
```
{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}
```

In[39]:=
```
(CosBisectAngSol //. Join[CosOrigAngSol] // Simplify) /. basicRule1
```
verknüpfe vereinfache

ReplaceAll: {basicRule1} is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

Out[39]=

$$\left\{\mathrm{Cos}[\mathrm{alphasab}] \to -\frac{(-1-2\,\mathrm{Cos}[a]+\mathrm{Cos}[2\,b]+2\,\mathrm{Cos}[b]\,\mathrm{Cos}[c])\,\mathrm{Csc}[b]\,\mathrm{Sec}\left[\frac{a}{2}\right]}{2\sqrt{-\frac{-2-4\,\mathrm{Cos}[a]+\mathrm{Cos}[2\,b]+4\,\mathrm{Cos}[b]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[a]}}},\right.$$

$$\mathrm{Cos}[\mathrm{alphasac}] \to \frac{\mathrm{Csc}[c]\,\mathrm{Sec}\left[\frac{a}{2}\right]\left(\mathrm{Cos}[a]-\mathrm{Cos}[b]\,\mathrm{Cos}[c]+\mathrm{Sin}[c]^2\right)}{\sqrt{-\frac{-2-4\,\mathrm{Cos}[a]+\mathrm{Cos}[2\,b]+4\,\mathrm{Cos}[b]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[a]}}},$$

$$\mathrm{Cos}[\mathrm{betasba}] \to -\frac{(-1+\mathrm{Cos}[2\,a]-2\,\mathrm{Cos}[b]+2\,\mathrm{Cos}[a]\,\mathrm{Cos}[c])\,\mathrm{Csc}[a]\,\mathrm{Sec}\left[\frac{b}{2}\right]}{2\sqrt{-\frac{-2+\mathrm{Cos}[2\,a]-4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[b]}}},$$

$$\mathrm{Cos}[\mathrm{betasbc}] \to \frac{\mathrm{Csc}[c]\,\mathrm{Sec}\left[\frac{b}{2}\right]\left(\mathrm{Cos}[b]-\mathrm{Cos}[a]\,\mathrm{Cos}[c]+\mathrm{Sin}[c]^2\right)}{\sqrt{-\frac{-2+\mathrm{Cos}[2\,a]-4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[b]}}},$$

$$\mathrm{Cos}[\mathrm{gammasca}] \to -\frac{(-1+\mathrm{Cos}[2\,a]+2\,\mathrm{Cos}[a]\,\mathrm{Cos}[b]-2\,\mathrm{Cos}[c])\,\mathrm{Csc}[a]\,\mathrm{Sec}\left[\frac{c}{2}\right]}{\sqrt{2}\sqrt{-\left((-2+\mathrm{Cos}[2\,a]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[b]+\mathrm{Cos}[2\,b]-4\,\mathrm{Cos}[c])\,\mathrm{Sec}\left[\frac{c}{2}\right]^2\right)}},$$

$$\mathrm{Cos}[\mathrm{gammascb}] \to -\frac{(-1+\mathrm{Cos}[a-b]+\mathrm{Cos}[2\,b]+\mathrm{Cos}[a+b]-2\,\mathrm{Cos}[c])\,\mathrm{Csc}[b]\,\mathrm{Sec}\left[\frac{c}{2}\right]}{\sqrt{2}\sqrt{-\left((-2+\mathrm{Cos}[2\,a]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[b]+\mathrm{Cos}[2\,b]-4\,\mathrm{Cos}[c])\,\mathrm{Sec}\left[\frac{c}{2}\right]^2\right)}},$$

$$\mathrm{Cos}[\mathrm{alphasbb}] \to \frac{(\mathrm{Cos}[a]-\mathrm{Cos}[c])\,\mathrm{Csc}\left[\frac{b}{2}\right]}{\sqrt{-\frac{-2+\mathrm{Cos}[2\,a]-4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[b]}}},$$

$$\mathrm{Cos}[\mathrm{alphascc}] \to \frac{\sqrt{2}\,(\mathrm{Cos}[a]-\mathrm{Cos}[b])\,\mathrm{Csc}\left[\frac{c}{2}\right]}{\sqrt{-\left((-2+\mathrm{Cos}[2\,a]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[b]+\mathrm{Cos}[2\,b]-4\,\mathrm{Cos}[c])\,\mathrm{Sec}\left[\frac{c}{2}\right]^2\right)}},$$

$$\mathrm{Cos}[\mathrm{betasaa}] \to \frac{(\mathrm{Cos}[b]-\mathrm{Cos}[c])\,\mathrm{Csc}\left[\frac{a}{2}\right]}{\sqrt{-\frac{-2-4\,\mathrm{Cos}[a]+\mathrm{Cos}[2\,b]+4\,\mathrm{Cos}[b]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[a]}}},$$

$$\mathrm{Cos}[\mathrm{betascc}] \to -\frac{\sqrt{2}\,(\mathrm{Cos}[a]-\mathrm{Cos}[b])\,\mathrm{Csc}\left[\frac{c}{2}\right]}{\sqrt{-\left((-2+\mathrm{Cos}[2\,a]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[b]+\mathrm{Cos}[2\,b]-4\,\mathrm{Cos}[c])\,\mathrm{Sec}\left[\frac{c}{2}\right]^2\right)}},$$

$$\mathrm{Cos}[\mathrm{gammasaa}] \to -\frac{(\mathrm{Cos}[b]-\mathrm{Cos}[c])\,\mathrm{Csc}\left[\frac{a}{2}\right]}{\sqrt{-\frac{-2-4\,\mathrm{Cos}[a]+\mathrm{Cos}[2\,b]+4\,\mathrm{Cos}[b]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[a]}}},$$

$$\left.\mathrm{Cos}[\mathrm{gammasbb}] \to \frac{(-\mathrm{Cos}[a]+\mathrm{Cos}[c])\,\mathrm{Csc}\left[\frac{b}{2}\right]}{\sqrt{-\frac{-2+\mathrm{Cos}[2\,a]-4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{1+\mathrm{Cos}[b]}}}\right\}\ /.\ \mathrm{basicRule1}$$

In[40]:=
```
% /. replaceVarRule // FullSimplify
```
vereinfache vollständig

ReplaceAll: {basicRule1} is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

Out[40]=

$$\left\{\text{Cosalphasab} \to \frac{2\left(\text{Cos}[a]-\text{Cos}[b]\,\text{Cos}[c]+\text{Sin}[b]^2\right)}{\sqrt{4-\text{ScaledSina2Sq}}\,\sqrt{\text{SinbSq}}\,\sqrt{-\frac{-2-4\,\text{Cos}[a]+\text{Cos}[2\,b]+4\,\text{Cos}[b]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[a]}}},\right.$$

$$\text{Cosalphasac} \to \frac{2\left(\text{SincSq}+\text{Cos}[a]-\text{Cos}[b]\,\text{Cos}[c]\right)}{\sqrt{4-\text{ScaledSina2Sq}}\,\sqrt{\text{SincSq}}\,\sqrt{-\frac{-2-4\,\text{Cos}[a]+\text{Cos}[2\,b]+4\,\text{Cos}[b]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[a]}}},$$

$$\text{Cosbetasba} \to -\frac{-1+\text{Cos}[2\,a]-2\,\text{Cos}[b]+2\,\text{Cos}[a]\,\text{Cos}[c]}{\sqrt{4-\text{ScaledSinb2Sq}}\,\sqrt{\text{SinaSq}}\,\sqrt{-\frac{-2+\text{Cos}[2\,a]-4\,\text{Cos}[b]+4\,\text{Cos}[a]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[b]}}},$$

$$\text{Cosbetasbc} \to \frac{2\left(\text{SincSq}+\text{Cos}[b]-\text{Cos}[a]\,\text{Cos}[c]\right)}{\sqrt{4-\text{ScaledSinb2Sq}}\,\sqrt{\text{SincSq}}\,\sqrt{-\frac{-2+\text{Cos}[2\,a]-4\,\text{Cos}[b]+4\,\text{Cos}[a]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[b]}}},$$

$$\text{Cosgammasca} \to \frac{1-\text{Cos}[2\,a]-2\,\text{Cos}[a]\,\text{Cos}[b]+2\,\text{Cos}[c]}{\sqrt{8-2\,\text{ScaledSinc2Sq}}\,\sqrt{\text{SinaSq}}\,\sqrt{\frac{-2+\text{Cos}[2\,a]+4\,\text{Cos}[a]\,\text{Cos}[b]+\text{Cos}[2\,b]-4\,\text{Cos}[c]}{-4+\text{ScaledSinc2Sq}}}},$$

$$\text{Cosgammascb} \to \frac{2\left(-\text{Cos}[a]\,\text{Cos}[b]+\text{Cos}[c]+\text{Sin}[b]^2\right)}{\sqrt{8-2\,\text{ScaledSinc2Sq}}\,\sqrt{\text{SinbSq}}\,\sqrt{\frac{-2+\text{Cos}[2\,a]+4\,\text{Cos}[a]\,\text{Cos}[b]+\text{Cos}[2\,b]-4\,\text{Cos}[c]}{-4+\text{ScaledSinc2Sq}}}},$$

$$\text{Cosalphasbb} \to \frac{2\left(\text{Cos}[a]-\text{Cos}[c]\right)}{\sqrt{\text{ScaledSinb2Sq}}\,\sqrt{-\frac{-2+\text{Cos}[2\,a]-4\,\text{Cos}[b]+4\,\text{Cos}[a]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[b]}}},$$

$$\text{Cosalphascc} \to \frac{\sqrt{2}\left(\text{Cos}[a]-\text{Cos}[b]\right)}{\sqrt{\text{ScaledSinc2Sq}}\,\sqrt{\frac{-2+\text{Cos}[2\,a]+4\,\text{Cos}[a]\,\text{Cos}[b]+\text{Cos}[2\,b]-4\,\text{Cos}[c]}{-4+\text{ScaledSinc2Sq}}}},$$

$$\text{Cosbetasaa} \to \frac{2\left(\text{Cos}[b]-\text{Cos}[c]\right)}{\sqrt{\text{ScaledSina2Sq}}\,\sqrt{-\frac{-2-4\,\text{Cos}[a]+\text{Cos}[2\,b]+4\,\text{Cos}[b]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[a]}}},$$

$$\text{Cosbetascc} \to \frac{\sqrt{2}\left(-\text{Cos}[a]+\text{Cos}[b]\right)}{\sqrt{\text{ScaledSinc2Sq}}\,\sqrt{\frac{-2+\text{Cos}[2\,a]+4\,\text{Cos}[a]\,\text{Cos}[b]+\text{Cos}[2\,b]-4\,\text{Cos}[c]}{-4+\text{ScaledSinc2Sq}}}},$$

$$\text{Cosgammasaa} \to -\frac{2\left(\text{Cos}[b]-\text{Cos}[c]\right)}{\sqrt{\text{ScaledSina2Sq}}\,\sqrt{-\frac{-2-4\,\text{Cos}[a]+\text{Cos}[2\,b]+4\,\text{Cos}[b]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[a]}}},$$

$$\left.\text{Cosgammasbb} \to \frac{2\left(-\text{Cos}[a]+\text{Cos}[c]\right)}{\sqrt{\text{ScaledSinb2Sq}}\,\sqrt{-\frac{-2+\text{Cos}[2\,a]-4\,\text{Cos}[b]+4\,\text{Cos}[a]\,\text{Cos}[c]+\text{Cos}[2\,c]}{1+\text{Cos}[b]}}}\right\}\ /.\ \text{basicRule1}$$

Confirm that alphasab + alphasac = alpha, betasba + betasbc = beta, gammasca + gammascb = gamma

In[41]:=
```
{Sin[b] Sin[c]
 Sinus   Sinus
    (Cos[alpha] - (Cos[alphasab] Cos[alphasac] - Sin[alphasab] Sin[alphasac])),
     Kosinus        Kosinus         Kosinus          Sinus          Sinus
   Sin[a] Sin[c] (Cos[beta] - (Cos[betasba] Cos[betasbc] - Sin[betasba] Sin[betasbc])),
   Sinus  Sinus   Kosinus       Kosinus       Kosinus        Sinus        Sinus
   Sin[a] Sin[b]
   Sinus  Sinus
    (Cos[gamma] - (Cos[gammasca] Cos[gammascb] - Sin[gammasca] Sin[gammascb]))
     Kosinus        Kosinus         Kosinus          Sinus          Sinus
  } //. Join[CosBisectAngSol, SinBisectAngSol,
        verknüpfe
   CosOrigAngSol, SinOrigAngSol, basicRule] // Simplify
                                               vereinfache
```

Out[41]=
```
{0, 0, 0}
```

Confirm that alphasbb + gammasbb = Pi, alphascc + betascc = Pi, betasaa + gammasaa = Pi

In[42]:=
```
Simplify[{
vereinfache
   Cos[alphasbb] Cos[gammasbb] - Sin[alphasbb] Sin[gammasbb],
   Kosinus       Kosinus         Sinus          Sinus
   Cos[alphascc] Cos[betascc] - Sin[alphascc] Sin[betascc],
   Kosinus       Kosinus        Sinus          Sinus
   Cos[betasaa] Cos[gammasaa] - Sin[betasaa] Sin[gammasaa]}
   Kosinus      Kosinus         Sinus          Sinus
  //. Join[CosBisectAngSol, SinBisectAngSol, CosOrigAngSol, SinOrigAngSol, basicRule]]
      verknüpfe
```

Out[42]=
```
{-1, -1, -1}
```

Formulas for angles within intersected triangles

In[43]:=
```
angleList = {
   Cos[alphasab], Cos[alphasac],
   Kosinus        Kosinus
   Cos[betasba], Cos[betasbc],
   Kosinus       Kosinus
   Cos[gammasca], Cos[gammascb],
   Kosinus        Kosinus
   Cos[alphasbb], Cos[alphascc],
   Kosinus        Kosinus
   Cos[betasaa], Cos[betascc],
   Kosinus       Kosinus
   Cos[gammasaa], Cos[gammasbb]};
   Kosinus        Kosinus
```

In[44]:=
```
tmp = ((Map[#^2 &, angleList] //. Join[CosBisectAngSol, SinBisectAngSol,
        wende an                   verknüpfe
            CosOrigAngSol, SinOrigAngSol, basicRule] // Simplify // ExpandAll //
                                                        vereinfache  multipliziere alle aus
        FullSimplify) //. basicRule) /. replaceVarRule // FullSimplify;
        vereinfache vollständig                          vereinfache vollständig
```

In[45]:=
```
tmp
```

Out[45]=

$$
\begin{aligned}
&\{(2\,\text{ScaledSina2Sq} + \text{ScaledSinb2Sq}\,(-2 + \text{ScaledSinc2Sq}) - 2\,(\text{ScaledSinc2Sq} + 2\,\text{SinbSq}))^2 / \\
&\quad (8\,\text{SinbSq}\,(-2\,\text{ScaledSina2Sq} - \text{ScaledSinb2Sq}\,(-2 + \text{ScaledSinc2Sq}) + \\
&\qquad 2\,(\text{ScaledSinc2Sq} + \text{SinbSq} + \text{SincSq}))), \\
&\quad -((2\,\text{ScaledSina2Sq} + \text{ScaledSinb2Sq}\,(-2 + \text{ScaledSinc2Sq}) - 2\,(\text{ScaledSinc2Sq} + 2\,\text{SincSq}))^2 / \\
&\qquad (8\,\text{SincSq}\,(2\,\text{ScaledSina2Sq} + \text{ScaledSinb2Sq}\,(-2 + \text{ScaledSinc2Sq}) - \\
&\qquad\quad 2\,(\text{ScaledSinc2Sq} + \text{SinbSq} + \text{SincSq})))), \\
&\quad (\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinc2Sq}) + 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} - 2\,\text{SinaSq}))^2 / \\
&\quad (8\,\text{SinaSq}\,(-\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinc2Sq}) + \\
&\qquad 2\,(-\text{ScaledSinb2Sq} + \text{ScaledSinc2Sq} + \text{SinaSq} + \text{SincSq}))), \\
&\quad (\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinc2Sq}) + 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} - 2\,\text{SincSq}))^2 / \\
&\quad (8\,\text{SincSq}\,(-\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinc2Sq}) + \\
&\qquad 2\,(-\text{ScaledSinb2Sq} + \text{ScaledSinc2Sq} + \text{SinaSq} + \text{SincSq}))), \\
&\quad (\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinb2Sq}) - 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} + 2\,\text{SinaSq}))^2 / \\
&\quad (8\,\text{SinaSq}\,(-\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinb2Sq}) + \\
&\qquad 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} + \text{SinaSq} + \text{SinbSq}))), \\
&\quad -((\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinb2Sq}) - 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} + 2\,\text{SinbSq}))^2 / \\
&\qquad (8\,\text{SinbSq}\,(\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinb2Sq}) - \\
&\qquad\quad 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} + \text{SinaSq} + \text{SinbSq})))), \\
&\quad ((-4 + \text{ScaledSinb2Sq})\,(\text{ScaledSina2Sq} - \text{ScaledSinc2Sq})^2) / \\
&\quad (2\,\text{ScaledSinb2Sq}\,(\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinc2Sq}) + \\
&\qquad 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} - \text{SinaSq} - \text{SincSq}))), \\
&\quad ((\text{ScaledSina2Sq} - \text{ScaledSinb2Sq})^2\,(-4 + \text{ScaledSinc2Sq})) / \\
&\quad (2\,\text{ScaledSinc2Sq}\,(\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinb2Sq}) - \\
&\qquad 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} + \text{SinaSq} + \text{SinbSq}))), \\
&\quad ((-4 + \text{ScaledSina2Sq})\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq})^2) / \\
&\quad (2\,\text{ScaledSina2Sq}\,(2\,\text{ScaledSina2Sq} + \\
&\qquad \text{ScaledSinb2Sq}\,(-2 + \text{ScaledSinc2Sq}) - 2\,(\text{ScaledSinc2Sq} + \text{SinbSq} + \text{SincSq}))), \\
&\quad ((\text{ScaledSina2Sq} - \text{ScaledSinb2Sq})^2\,(-4 + \text{ScaledSinc2Sq})) / \\
&\quad (2\,\text{ScaledSinc2Sq}\,(\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinb2Sq}) - \\
&\qquad 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} + \text{SinaSq} + \text{SinbSq}))), \\
&\quad ((-4 + \text{ScaledSina2Sq})\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq})^2) / \\
&\quad (2\,\text{ScaledSina2Sq}\,(2\,\text{ScaledSina2Sq} + \\
&\qquad \text{ScaledSinb2Sq}\,(-2 + \text{ScaledSinc2Sq}) - 2\,(\text{ScaledSinc2Sq} + \text{SinbSq} + \text{SincSq}))), \\
&\quad ((-4 + \text{ScaledSinb2Sq})\,(\text{ScaledSina2Sq} - \text{ScaledSinc2Sq})^2) / \\
&\quad (2\,\text{ScaledSinb2Sq}\,(\text{ScaledSina2Sq}\,(-2 + \text{ScaledSinc2Sq}) + \\
&\qquad 2\,(\text{ScaledSinb2Sq} - \text{ScaledSinc2Sq} - \text{SinaSq} - \text{SincSq})))\}
\end{aligned}
$$

In[46]:=
```
angleFormulas = Join[
                verknüpfe
   FullSimplify[Map[
   vereinfache voll··· wende an
     ((angleList[[#]] /. replaceVarRule) → Sqrt[tmp[[#]]]) &, Range[Length[angleList]]]],
                                          Quadratwurzel        Liste ··· Länge
   FullSimplify[
   vereinfache vollständig
    Map[((angleList[[#]]^2 /. replaceVarRule) → tmp[[#]]) &, Range[Length[angleList]]]]];
    wende an                                                  Liste ··· Länge
```

In[47]:=
```
formulas = Join[formulas, angleFormulas];
           verknüpfe
```

Distances center point to corners and edges: da, db, dc, ea, eb, ec

Use spherical sine law to calculate the ratios (also shows that the three side bisections intersect in one point, since the ratios only depend on the bisected side itself):
Sin[da] = 2 Cos[a/2] Sin[ea], Sin[db] = 2 Cos[b/2] Sin[eb], Sin[dc] = 2 Cos[c/2] Sin[ec]

In[48]:=
```
cornerDistRatioSol = Solve[
                     löse
     {SpherSinLaw[a / 2, lambda, ea, gammasca],
      SpherSinLaw[c / 2, lambda, da, betascc],
      SpherSinLaw[b, betascc, c / 2, gammascb],
      SpherSinLaw[a, b, gammascb, gammasca],

      SpherSinLaw[b / 2, mu, eb, alphasab],
      SpherSinLaw[a / 2, mu, db, gammasaa],
      SpherSinLaw[c, gammasaa, a / 2, alphasac],
      SpherSinLaw[b, c, alphasac, alphasab],

      SpherSinLaw[c / 2, nu, ec, betasbc],
      SpherSinLaw[b / 2, nu, dc, alphasbb],
      SpherSinLaw[a, alphasbb, b / 2, betasba],
      SpherSinLaw[a, c, betasbc, betasba]},
     {Sin[da], Sin[db], Sin[dc], Sin[gammascb],
      Sinus     Sinus    Sinus    Sinus
      Sin[alphasac], Sin[betasba], Sin[betascc], Sin[gammasaa],
      Sinus          Sinus         Sinus         Sinus
      Sin[alphasbb], Sin[lambda], Sin[mu], Sin[nu]}][[1]] // Simplify;
      Sinus          Sinus        Sinus    Sinus          vereinfache
```

Calculate ea, eb, ec through ea+da=sa, eb+db=sb, ec+dc=sc

In[49]:=
```
sadaeaEquations = {(Cos[sa] + Sin[da] Sin[ea])^2 == (1 - Sin[da]^2) (1 - Sin[ea]^2),
                   Kosinus  Sinus  Sinus          Sinus            Sinus
   (Cos[sb] + Sin[db] Sin[eb])^2 == (1 - Sin[db]^2) (1 - Sin[eb]^2),
    Kosinus  Sinus  Sinus          Sinus            Sinus
   (Cos[sc] + Sin[dc] Sin[ec])^2 == (1 - Sin[dc]^2) (1 - Sin[ec]^2)
    Kosinus  Sinus  Sinus          Sinus            Sinus
  };
```

Check it these equations represent sa = da + ea

In[50]:=
```
Simplify[sadaeaEquations /. {sa → da + ea, sb → db + eb, sc → dc + ec}]
vereinfache
```

Out[50]=
```
{True, True, True}
```

In[51]:=
```
eaSol = (Solve[(sadaeaEquations /. cornerDistRatioSol) /. sdeVarNameRule,
         löse
          {SineaSq, SinebSq, SinecSq}][[1]] // ExpandAll // Simplify) /.
                                               multipliziere all··· vereinfache
   (sdeVarNameRule /. {Rule[a_, b_] → Rule[b, a]});
                       Regel          Regel
```

In[52]:=
```
eaFormula =
  FullSimplify[eaSol //. Join[SinSSol, CosSSol, CosOrigAngSol, basicRule] // ExpandAll,
  vereinfache vollständig  verknüpfe                                        multipliziere all
   {a ≥ 0, a ≤ Pi / 2}];
              Kreiszahl π
```

Check that the results are unchanged when swapping variable names

In[53]:=
```
{(Sin[ea]^2 /. eaFormula) - ((Sin[eb]^2 /. eaFormula) /. {a → b, b → a}),
  Sinus                      Sinus
 (Sin[ea]^2 /. eaFormula) - ((Sin[ec]^2 /. eaFormula) /. {a → c, c → a}),
  Sinus                      Sinus
 (Sin[eb]^2 /. eaFormula) - ((Sin[ec]^2 /. eaFormula) /. {b → c, c → b})} // Simplify
  Sinus                      Sinus                                        vereinfache
```

Out[53]=
```
{0, 0, 0}
```

In[54]:=
```
eaFormula = eaFormula //. Join[basicRule, replaceVarRule, sdeVarNameRule] // FullSimplify;
                          verknüpfe                                        vereinfache vollstär
```

In[55]:=
```
formulas = Join[formulas, eaFormula];
           verknüpfe
```

In[56]:=
```
daFormula =
  (Map[(# → (# /. cornerDistRatioSol) //. Join[eaSol, SinSSol, CosSSol, CosOrigAngSol]) &,
   {Sin[da]^2, Sin[db]^2, Sin[dc]^2}] // FullSimplify)
```
wende an · verknüpfe · Sinus · Sinus · Sinus · vereinfache vollständig

Out[56]=

$$\left\{\mathrm{Sin}[da]^2 \to -\frac{-2-4\,\mathrm{Cos}[a]+\mathrm{Cos}[2\,b]+4\,\mathrm{Cos}[b]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{6+4\,\mathrm{Cos}[a]+4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[c]},\right.$$

$$\mathrm{Sin}[db]^2 \to -\frac{-2+\mathrm{Cos}[2\,a]-4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[c]+\mathrm{Cos}[2\,c]}{6+4\,\mathrm{Cos}[a]+4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[c]},$$

$$\left.\mathrm{Sin}[dc]^2 \to -\frac{-2+\mathrm{Cos}[2\,a]+4\,\mathrm{Cos}[a]\,\mathrm{Cos}[b]+\mathrm{Cos}[2\,b]-4\,\mathrm{Cos}[c]}{6+4\,\mathrm{Cos}[a]+4\,\mathrm{Cos}[b]+4\,\mathrm{Cos}[c]}\right\}$$

In[57]:=
```
daFormula = daFormula //. Join[sdeVarNameRule, basicRule, replaceVarRule] // Simplify;
```
verknüpfe · vereinfache

In[58]:=
```
formulas = Join[formulas, daFormula];
```
verknüpfe

Final results

In[59]:=
```
MatrixForm[formulas]
```
Matritzenform

Out[59]//MatrixForm=

$$\mathrm{ScaledSina2Sq} \to \frac{2\,\mathrm{SinaSq}}{1+\sqrt{1-\mathrm{SinaSq}}}$$

$$\mathrm{ScaledSinb2Sq} \to \frac{2\,\mathrm{SinbSq}}{1+\sqrt{1-\mathrm{SinbSq}}}$$

$$\mathrm{ScaledSinc2Sq} \to \frac{2\,\mathrm{SincSq}}{1+\sqrt{1-\mathrm{SincSq}}}$$

$$\mathrm{CosAlpha} \to \frac{-2\,\mathrm{ScaledSina2Sq}-\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})+2\,\mathrm{ScaledSinc2Sq}}{4\,\sqrt{\mathrm{SinbSq}}\,\sqrt{\mathrm{SincSq}}}$$

$$\mathrm{CosBeta} \to \frac{-2\,\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinc2Sq})+2\,\mathrm{ScaledSinc2Sq}}{4\,\sqrt{\mathrm{SinaSq}}\,\sqrt{\mathrm{SincSq}}}$$

$$\mathrm{CosGamma} \to \frac{2\,\mathrm{ScaledSina2Sq}+2\,\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSina2Sq}\,\mathrm{ScaledSinb2Sq}-2\,\mathrm{ScaledSinc2Sq}}{4\,\sqrt{\mathrm{SinaSq}}\,\sqrt{\mathrm{SinbSq}}}$$

$$\mathrm{SinsaSq} \to \frac{2\,\mathrm{ScaledSina2Sq}+\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})-2\,(\mathrm{ScaledSinc2Sq}+\mathrm{SinbSq}+\mathrm{SincSq})}{2\,(-4+\mathrm{ScaledSina2Sq})}$$

$$\mathrm{SinsbSq} \to \frac{-\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinc2Sq})+2\,(-\mathrm{ScaledSinb2Sq}+\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}+\mathrm{SincSq})}{8-2\,\mathrm{ScaledSinb2Sq}}$$

$$\mathrm{SinscSq} \to \frac{\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinb2Sq})-2\,(\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}+\mathrm{SinbSq})}{2\,(-4+\mathrm{ScaledSinc2Sq})}$$

$$\mathrm{SinmaSq} \to \frac{\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})-\mathrm{ScaledSina2Sq}\,(-6+\mathrm{ScaledSinb2Sq}+\mathrm{ScaledSinc2Sq})+2\,(-\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}+\mathrm{S}}{2\,(-4+\mathrm{ScaledSinb2Sq})\,(-4+\mathrm{ScaledSinc2Sq})}$$

$$\mathrm{SinmbSq} \to \frac{-\mathrm{ScaledSinb2Sq}\,(-6+\mathrm{ScaledSinc2Sq})+\mathrm{ScaledSina2Sq}\,(-2-\mathrm{ScaledSinb2Sq}+\mathrm{ScaledSinc2Sq})+2\,(-\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}+\mathrm{S}}{2\,(-4+\mathrm{ScaledSina2Sq})\,(-4+\mathrm{ScaledSinc2Sq})}$$

$$\mathrm{SinmcSq} \to \frac{\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSinc2Sq})-\mathrm{ScaledSinb2Sq}\,(2+\mathrm{ScaledSinc2Sq})+2\,(3\,\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}+\mathrm{S}}{2\,(-4+\mathrm{ScaledSina2Sq})\,(-4+\mathrm{ScaledSinb2Sq})}$$

$$4\,\mathrm{SinmaSq} \to \frac{2\,\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})-2\,\mathrm{ScaledSina2Sq}\,(-6+\mathrm{ScaledSinb2Sq}+\mathrm{ScaledSinc2Sq})+4\,(-\mathrm{ScaledSinc2Sq}+\mathrm{SinaS}}{(-4+\mathrm{ScaledSinb2Sq})\,(-4+\mathrm{ScaledSinc2Sq})}$$

$$4\,\mathrm{SinmbSq} \to \frac{-2\,\mathrm{ScaledSina2Sq}\,(2+\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSinc2Sq})-2\,\mathrm{ScaledSinb2Sq}\,(-6+\mathrm{ScaledSinc2Sq})+4\,(-\mathrm{ScaledSinc2Sq}+\mathrm{SinaS}}{(-4+\mathrm{ScaledSina2Sq})\,(-4+\mathrm{ScaledSinc2Sq})}$$

$$4\,\mathrm{SinmcSq} \to \frac{2\,\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSinc2Sq})-2\,\mathrm{ScaledSinb2Sq}\,(2+\mathrm{ScaledSinc2Sq})+4\,(3\,\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}}{(-4+\mathrm{ScaledSina2Sq})\,(-4+\mathrm{ScaledSinb2Sq})}$$

$$\mathrm{Cosalphasab} \to \frac{\sqrt{\frac{(2\,\mathrm{ScaledSina2Sq}+\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})-2\,(\mathrm{ScaledSinc2Sq}+2\,\mathrm{SinbSq}))^2}{\mathrm{SinbSq}\,(-2\,\mathrm{ScaledSina2Sq}-\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})+2\,(\mathrm{ScaledSinc2Sq}+\mathrm{SinbSq}+\mathrm{SincSq}))}}}{2\,\sqrt{2}}$$

$$\mathrm{Cosalphasac} \to \frac{\sqrt{-\frac{(2\,\mathrm{ScaledSina2Sq}+\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})-2\,(\mathrm{ScaledSinc2Sq}+2\,\mathrm{SincSq}))^2}{\mathrm{SincSq}\,(2\,\mathrm{ScaledSina2Sq}+\mathrm{ScaledSinb2Sq}\,(-2+\mathrm{ScaledSinc2Sq})-2\,(\mathrm{ScaledSinc2Sq}+\mathrm{SinbSq}+\mathrm{SincSq}))}}}{2\,\sqrt{2}}$$

$$\mathrm{Cosbetasba} \to \frac{1}{2}\,\sqrt{\frac{(\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinc2Sq})+2\,(\mathrm{ScaledSinb2Sq}-\mathrm{ScaledSinc2Sq}-2\,\mathrm{SinaSq}))^2}{-2\,\mathrm{ScaledSina2Sq}\,(-2+\mathrm{ScaledSinc2Sq})\,\mathrm{SinaSq}+4\,\mathrm{SinaSq}\,(-\mathrm{ScaledSinb2Sq}+\mathrm{ScaledSinc2Sq}+\mathrm{SinaSq}+\mathrm{Si}}}$$

$$\text{Cosbetasbc} \to \frac{\sqrt{\frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}-2\,\text{SincSq}))^2}{\text{SincSq}\,(-\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(-\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SincSq}))}}}{2\sqrt{2}}$$

$$\text{Cosgammasca} \to \frac{1}{2}\sqrt{\frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+2\,\text{SinaSq}))^2}{-2\,\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})\,\text{SinaSq}+4\,\text{SinaSq}\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{Si}}}$$

$$\text{Cosgammascb} \to \frac{1}{2}\sqrt{\frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+2\,\text{SinbSq}))^2}{-2\,\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})\,\text{SinbSq}+4\,\text{SinbSq}\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{Si}}}$$

$$\text{Cosalphasbb} \to \frac{\sqrt{\frac{(-4+\text{ScaledSinb2Sq})\,(\text{ScaledSina2Sq}-\text{ScaledSinc2Sq})^2}{\text{ScaledSinb2Sq}\,(-2\,\text{ScaledSina2Sq}+2\,\text{ScaledSinb2Sq}-2\,\text{ScaledSinc2Sq}+\text{ScaledSina2Sq}\,\text{ScaledSinc2Sq}-2\,\text{SinaSq}-2\,\text{SincSq})}}}{\sqrt{2}}$$

$$\text{Cosalphascc} \to \frac{\sqrt{\frac{(\text{ScaledSina2Sq}-\text{ScaledSinb2Sq})^2\,(-4+\text{ScaledSinc2Sq})}{\text{ScaledSinc2Sq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SinbSq}))}}}{\sqrt{2}}$$

$$\text{Cosbetasaa} \to \frac{\sqrt{\frac{(-4+\text{ScaledSina2Sq})\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq})^2}{\text{ScaledSina2Sq}\,(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{SincSq}))}}}{\sqrt{2}}$$

$$\text{Cosbetascc} \to \frac{\sqrt{\frac{(\text{ScaledSina2Sq}-\text{ScaledSinb2Sq})^2\,(-4+\text{ScaledSinc2Sq})}{\text{ScaledSinc2Sq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SinbSq}))}}}{\sqrt{2}}$$

$$\text{Cosgammasaa} \to \frac{\sqrt{\frac{(-4+\text{ScaledSina2Sq})\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq})^2}{\text{ScaledSina2Sq}\,(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{SincSq}))}}}{\sqrt{2}}$$

$$\text{Cosgammasbb} \to \frac{\sqrt{\frac{(-4+\text{ScaledSinb2Sq})\,(\text{ScaledSina2Sq}-\text{ScaledSinc2Sq})^2}{\text{ScaledSinb2Sq}\,(-2\,\text{ScaledSina2Sq}+2\,\text{ScaledSinb2Sq}-2\,\text{ScaledSinc2Sq}+\text{ScaledSina2Sq}\,\text{ScaledSinc2Sq}-2\,\text{SinaSq}-2\,\text{SincSq})}}}{\sqrt{2}}$$

$$\text{Cosalphasab}^2 \to \frac{(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+2\,\text{SinbSq}))^2}{8\,\text{SinbSq}\,(-2\,\text{ScaledSina2Sq}-\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{SincSq}}$$

$$\text{Cosalphasac}^2 \to -\frac{(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+2\,\text{SincSq}))^2}{8\,\text{SincSq}\,(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{SincSc}}$$

$$\text{Cosbetasba}^2 \to \frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}-2\,\text{SinaSq}))^2}{8\,\text{SinaSq}\,(-\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(-\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SincSq})}$$

$$\text{Cosbetasbc}^2 \to \frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}-2\,\text{SincSq}))^2}{8\,\text{SincSq}\,(-\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(-\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SincSq})}$$

$$\text{Cosgammasca}^2 \to \frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+2\,\text{SinaSq}))^2}{8\,\text{SinaSq}\,(-\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SinbSq})}$$

$$\text{Cosgammascb}^2 \to -\frac{(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+2\,\text{SinbSq}))^2}{8\,\text{SinbSq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SinbSq}}$$

$$\text{Cosalphasbb}^2 \to \frac{(-4+\text{ScaledSinb2Sq})\,(\text{ScaledSina2Sq}-\text{ScaledSinc2Sq})^2}{2\,\text{ScaledSinb2Sq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}-\text{SinaSq}-\text{Sinc}}$$

$$\text{Cosalphascc}^2 \to \frac{(\text{ScaledSina2Sq}-\text{ScaledSinb2Sq})^2\,(-4+\text{ScaledSinc2Sq})}{2\,\text{ScaledSinc2Sq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{Sinb}}$$

$$\text{Cosbetasaa}^2 \to \frac{(-4+\text{ScaledSina2Sq})\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq})^2}{2\,\text{ScaledSina2Sq}\,(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{Sinc}}$$

$$\text{Cosbetascc}^2 \to \frac{(\text{ScaledSina2Sq}-\text{ScaledSinb2Sq})^2\,(-4+\text{ScaledSinc2Sq})}{2\,\text{ScaledSinc2Sq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{Sinb}}$$

$$\text{Cosgammasaa}^2 \to \frac{(-4+\text{ScaledSina2Sq})\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq})^2}{2\,\text{ScaledSina2Sq}\,(2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{Sin}}$$

$$\text{Cosgammasbb}^2 \to \frac{(-4+\text{ScaledSinb2Sq})\,(\text{ScaledSina2Sq}-\text{ScaledSinc2Sq})^2}{2\,\text{ScaledSinb2Sq}\,(\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}-\text{SinaSq}-\text{Sinc}}$$

$$\text{SineaSq} \to -\frac{2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{SincSq})}{2\,(-4+\text{ScaledSina2Sq})\,(-9+\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq})}$$

$$\text{SinebSq} \to \frac{-\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(-\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SincSq})}{2\,(-4+\text{ScaledSinb2Sq})\,(-9+\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq})}$$

$$\text{SinecSq} \to \frac{-\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SinbSq})}{2\,(-4+\text{ScaledSinc2Sq})\,(-9+\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq})}$$

$$\text{SindaSq} \to \frac{2\,\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}\,(-2+\text{ScaledSinc2Sq})-2\,(\text{ScaledSinc2Sq}+\text{SinbSq}+\text{SincSq})}{2\,(-9+\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq})}$$

$$\text{SindbSq} \to \frac{\text{ScaledSina2Sq}\,(-2+\text{ScaledSinc2Sq})+2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}-\text{SinaSq}-\text{SincSq})}{2\,(-9+\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq})}$$

$$\text{SindcSq} \to \frac{\text{ScaledSina2Sq}\,(-2+\text{ScaledSinb2Sq})-2\,(\text{ScaledSinb2Sq}-\text{ScaledSinc2Sq}+\text{SinaSq}+\text{SinbSq})}{2\,(-9+\text{ScaledSina2Sq}+\text{ScaledSinb2Sq}+\text{ScaledSinc2Sq})}$$

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